

Classifying All Groups of Order 16

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1 Introduction

In group theory, we are often interested in classifying all groups of a certain order by isomorphism class, which demonstrates that they have the same structure. For orders up to 15, we have already determined the isomorphism classes. This paper will extend our classification to the groups of order 16. To begin, we introduce some basic notation:

Notation *A group of order 16 will be denoted G . The symbol $\cong$ will stand for 'isomorphic'.*

In order to describe any group, the representation $G = \{a_1^{\alpha_1} a_2^{\alpha_2} \dots a_n^{\alpha_n} : a_1^{\beta_1} = a_2^{\beta_2} = \dots a_n^{\beta_n} = e, a_2 a_1 = a_1 a_2 a_{1,2}, a_3 a_1 = a_1 a_3 a_{1,3}, \dots a_n a_{n-1} = a_{n-1} a_n a_{n-1,n}\}$ will be used. This tells us the elements of G in terms of generators and the orders of the generators. The representation also describes how the generators commute, so we can condense a string of the elements into the form used in the presentation. Two notes: I have left out a piece where, occasionally, powers of the generators equal each other when they are less than their orders (which would also be stated). Also, you cannot have just any old representation, but that is a different paper topic. We begin by noting that the prime factorization of 16 is $16 = 2^4$, so any group of order 16 is a p -group (a group whose order is the power of a prime, in this case 2). As such we will classify with factor groups. In particular, we will use the center of a group in our classification:

Definition *The center of any group is the set of all elements that commute with every element in a group, denoted $Z(G) = \{z : zg = gz, \forall g \in G\}$.*

We have the following two theorems about the center:

Theorem 1.1 *For a p -group, The center of a group is a nontrivial subgroup (Judson, 186).*

Theorem 1.2 *The center of a group is also a normal subgroup (Judson, 186).*

To classify the groups of order 16, we consider the different cases for the order of the center. Since $Z(G)$ is a nontrivial subgroup, $Z(G)$ must divide the order of the group, so the possibilities for $|Z(G)|$ are 16, 8, 4, and 2. Based on the center, we then build the factor group $G/Z(G)$, which will have order $|G/Z(G)| = |G|/|Z(G)|$. Then using the Correspondence Theorem, we deduce the properties of the group for various cases.

Theorem 1.3 (Correspondence Theorem) *If N is a normal subgroup of any group G . Then $S \mapsto S/N$ is a one-to-one correspondence between the set of subgroups S containing N and the subgroups of G/N . Furthermore, the normal subgroups in S correspond to normal subgroups in G/N , and if a subgroup of G/N is contained in a subgroup of G/N , then the corresponding subgroups in S share the same relation (Judson, 147).*

From this, we see that we need to know the groups of order 8, 4, and 2, shown in the table below.

Name	Order	Symbol	Representation	Number and Structure of Non-trivial Subgroups	Center
Integers mod 2	2	$\mathbb{Z}_2$	$\{a^\alpha : a^2 = e\}$	None	abelian

Name	Order	Symbol	Representation	Number and Structure of Non-trivial Subgroups	Center
Integers mod 4	4	$\mathbb{Z}_4$	$\{a^\alpha : a^4 = e\}$	1 isomorphic to $\mathbb{Z}_2$	abelian
Klein 4 group	4	$\mathbb{Z}_2 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta : a^2 = b^2 = e, ba = ab\}$	3 isomorphic to $\mathbb{Z}_2$	abelian
Integers mod 8	8	$\mathbb{Z}_8$	$\{a^\alpha : a^8 = e\}$	1 isomorphic to $\mathbb{Z}_4$ 1 isomorphic to $\mathbb{Z}_2$	abelian
Direct Product of $\mathbb{Z}_4$ and $\mathbb{Z}_2$	8	$\mathbb{Z}_4 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta : a^4 = b^2 = e, ba = ab\}$	2 isomorphic to $\mathbb{Z}_4$ 1 isomorphic to $\mathbb{Z}_2$ $\times \mathbb{Z}_2$, 3 isomorphic to $\mathbb{Z}_2$	abelian
Direct Product of $\mathbb{Z}_2$, $\mathbb{Z}_2$ and $\mathbb{Z}_2$	8	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta c^\gamma : a^2 = b^2 = c^2 = e, ba = ab, ca = ac, cb = bc\}$	7 isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$, 7 isomorphic to $\mathbb{Z}_2$	abelian
Quaternion Group	8	$\mathcal{Q}$	$\{a^\alpha b^\beta : a^2 = b^2, a^4 = b^4 = e, ba = a^{-1}b\}$	3 isomorphic to $\mathbb{Z}_4$ 1 isomorphic to $\mathbb{Z}_2$	$\{e, a^2\} \cong \mathbb{Z}_2$
Dihedral Group	8	D_4	$\{a^\alpha b^\beta : a^4 = b^2 = e, ba = a^{-1}b\}$	1 isomorphic to $\mathbb{Z}_4$ 2 isomorphic to $\mathbb{Z}_2$ $\times \mathbb{Z}_2$, 5 isomorphic to $\mathbb{Z}_2$	$\{e, a^2\} \cong \mathbb{Z}_2$

Table 1: Groups of Order 2, 4, and 8 (Source: sagenb.org)

2 $|Z(G)| = 16$

To begin, assume $|Z(G)| = 16 = |G|$. Since a subset of a finite set equals the set if they have the same number of elements, $Z(G) = G$. Next, we note a fairly obvious fact:

Theorem 2.1 *A group is abelian if and only if $Z(G) = G$*

Proof: If a group is abelian, then every element commutes with every element, so every element is in the center. Likewise, if the center equals the group, then every element is in the center and commutes with every element, and thus the group is abelian.

Since our group is abelian, we can use the Fundamental Theorem of Abelian Groups:

Theorem 2.2 (Fundamental Theorem of Finite Abelian Groups) *Every finite abelian group is isomorphic to a direct product of cyclic groups of the form $\mathbb{Z}_{p_1^{\alpha_1}} \times \mathbb{Z}_{p_2^{\alpha_2}} \times \dots \times \mathbb{Z}_{p_n^{\alpha_n}}$, where the p_i are (not necessarily distinct) primes (Judson, 172).*

Since the group is isomorphic to the direct product of cyclic groups, we note that the only possibilities for the order of cyclic groups are powers of 2. The sum of the powers must equal 4, so we have 5 ways of writing 4 as the sum of positive integers: $4=4$, $4=3+1$, $4=2+2$, $4=2+1+1$, and $4=1+1+1+1$. Thus, there are five abelian isomorphism classes for the groups of order sixteen,

$$G \cong \mathbb{Z}_{2^4} = \mathbb{Z}_{16}, \quad G \cong \mathbb{Z}_{2^3} \times \mathbb{Z}_2 = \mathbb{Z}_8 \times \mathbb{Z}_2, \quad G \cong \mathbb{Z}_{2^2} \times \mathbb{Z}_{2^2} = \mathbb{Z}_4 \times \mathbb{Z}_4, \\ G \cong \mathbb{Z}_{2^2} \times \mathbb{Z}_2 \times \mathbb{Z}_2 = \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2, \quad G \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$$

We will put these in the table below, with their representations.

Name	Symbol	Representation	Center
Integers mod 16	$\mathbb{Z}_{16}$	$\{a^\alpha : a^{16} = e\}$	abelian
Direct Product of $\mathbb{Z}_8$ and $\mathbb{Z}_2$	$\mathbb{Z}_8 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta : a^8 = b^2 = e, ba = ab\}$	abelian
Direct Product of $\mathbb{Z}_4$ and $\mathbb{Z}_4$	$\mathbb{Z}_4 \times \mathbb{Z}_4$	$\{a^\alpha b^\beta : a^4 = b^4 = e, ba = ab\}$	abelian
Direct Product of $\mathbb{Z}_4$, $\mathbb{Z}_2$ and $\mathbb{Z}_2$	$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta c^\gamma : a^4 = b^2 = c^2 = e, ba = ab, ca = ac, cb = bc\}$	abelian
Direct product of $\mathbb{Z}_2$, $\mathbb{Z}_2$, $\mathbb{Z}_2$ and $\mathbb{Z}_2$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta c^\gamma d^\delta : a^2 = b^2 = c^2 = d^2 = e, ba = ab, ca = ac, da = ad, cb = bc, db = bd, dc = cd\}$	abelian

Table 2: Abelian Groups of Order 16

3 Using the Correspondence Theorem

From this point on, I will follow a basic method for determining the non-abelian groups. I will start by stating the order of the center, and then from this state the possibilities for the group based on the structure of the center and factor group. In doing so, I hope to either determine if a contradiction occurs, or find a representation for the group based on the elements needed to generate the group and how they commute. I will thus use the following notation and theorems, which apply regardless of the order of the center (any conditions for these theorems will be stated). To begin, we note that, as the center is a normal subgroup, the factor group will have subgroups. By the Correspondence Theorem, these subgroups contain the center, and some of these will be of order 8. We denote these subgroups, and prove a few theorems about them, below:

Notation Denote the subgroups of order 8 in G containing $Z(G)$ by G_i where i indexes the subgroup. Also, unless specified z will denote an element of $Z(G)$, and g_i will denote a element of G_i not in $Z(G)$.

Theorem 3.1 The center of G_i contains the center of G , $Z(G) \subset Z(G_i)$.

Proof: This is trivial as an element is in the center commutes with every element, and thus commutes with all of G_i .

Next we state a useful theorem about the product set of two groups, one which we will use repeatedly:

Theorem 3.2 Let H, K be subgroups of G . Then the subset $HK = \{hk : h \in H, k \in K\}$ has order $|HK| = |H||K|/|H \cap K|$ (Judson, 203).

From this, we see some useful facts about the G_i :

Theorem 3.3 The intersection of two G_i must be a group of order 4.

Proof: Let G_i, G_j be distinct subgroups of order 8 containing the center. Then, by theorem 4.2, $|G_i G_j| = |G_i||G_j|/|G_i \cap G_j| = 8 * 8 / |G_i \cap G_j| = 64 / |G_i \cap G_j|$. Since the intersection of a subgroup is a subgroup (Judson, 46), the order of $G_i \cap G_j$ must be either 8, 4, 2, or 1. if $|G_i \cap G_j| = 8$, then the groups are the same, and for order 1 or 2, we get that $|G_i G_j|$ is 64 and 32, respectively. This leads to a contradiction, as then $G_i G_j$ is larger than G . Thus, the order of the intersection of two distinct G_i is a group of order 4.

Theorem 3.4 *Three distinct G_i that share a common intersection (that will be a group of order 4 by Theorem 4.3) are formed by the cosets of their intersection. In particular $G_i = (G_i \cap G_j) \cup g_i(G_i \cap G_j)$, where g_i is not in the intersection. If three distinct G_i share a common intersection, then they contain every element in the group*

Proof: To show that this is true, note that both $(G_i \cap G_j), g_i(G_i \cap G_j)$ are subsets of G_i , and since g_i is not in the intersection, they are disjoint. Hence there are eight elements in the union, which equals the number of elements in G_i , so the sets are equal. From this, having three G_i means we have four distinct cosets (one that is the intersection and three for each G_i) so these are the number of cosets, and every element is in a coset by Lagrange's Theorem (Judson, 81), so every element is in at least one G_i (as every element is in a coset of the intersection as a subgroup).

Theorem 3.5 *Let g_i, g_j be elements of G_i, G_j with neither of them in $G_i \cap G_j$. Then $g_i g_j$ is not in G_i or G_j .*

Proof: Without loss of generality, we will show $g_i g_j \notin G_i$. For the purposes of contradiction, assume that for $h_i \in G_i$, $h_i = g_i g_j$. Then $g_j = g_i^{-1} h_i$ is in G_i , which would mean g_j is in the intersection of G_i and G_j , which contradicts our assumptions about g_j . So $g_i g_j$ is not in either G_i or G_j .

Theorem 3.6 *Let $G/Z(G)$ be abelian. Then the commutator between g_i, g_j is an element z' of the center, $g_i g_j = z' g_j g_i$.*

Proof: let $g_i Z(G), g_j Z(G) \in G/Z(G)$. Then since the factor group is abelian, $(g_i Z(G))(g_j Z(G)) = (g_j Z(G))(g_i Z(G))$. Note that when we perform the operation in the factor group, we get $g_i g_j Z(G) = g_j g_i Z(G)$, so $g_i g_j = z' g_j g_i$ for some $z' \in Z(G)$ from the properties of cosets.

4 $|Z(G)| = 8$

In the case that $|Z(G)| = 8$, then we form the factor group $G/Z(G)$, which has order $|G/Z(G)| = |G|/|Z(G)| = 16/8 = 2$. Since the factor group is a group of order 2, the factor group is cyclic. However, we have a theorem about the factor group being cyclic:

Theorem 4.1 *If $G/Z(G)$ is cyclic, then G is abelian (Judson, 186).*

By theorem 2.1, a group that is abelian has a center equal to the group. If the center equals the group, then the order of the center is 16 and not 8. Hence we have a contradiction, so no groups of order 16 have a center of order 8.

5 $|Z(G)| = 4$

If the order of the center is four, then the order of $G/Z(G)$ is $|G/Z(G)| = |G|/|Z(G)| = 16/4 = 4$. The only possibilities for the factor group are thus $G/Z(G) \cong \mathbb{Z}_4$ or $G/Z(G) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. If $G/Z(G) \cong \mathbb{Z}_4$, then the factor group is cyclic, which as we saw above implies a center of order sixteen, not four. Hence the factor group must be isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. We note that, in this case, there are three subgroups of order 2, and by the Correspondence Theorem, there are three G_i (the mapping from G to $G/Z(G)$ is a 4 to 1 map, hence a subgroup of order 2 in the factor group is of order $2 * 4 = 8$ in G). We also note that by Theorem 3.3, the intersection of two G_i is the center, as it is in the intersection and a subgroup of order 4.

We next note that, with $|Z(G)| = 4$ the only possibilities are that all G_i are abelian, since by theorem 3.1, the center of G_i , would have at least four elements in $Z(G_i)$, and not two as with the non-abelian groups of order 8. Hence G_i are $\cong \mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, or $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. We now note a fact about commuting two elements

Theorem 5.1 *If $|Z(G)| = 4$, and g_i, g_j with $i \neq j$, then they do not commute.*

Proof: If g_i, g_j commute when they are in different G_i , then g_i commutes with all elements in $Z(G)$ (as the center), $g_i Z(G)$ (as commutes with itself and the center) $g_j Z(G)$ (as this is g_j times an element of the center). This means there are at least 12 elements in the centralizer of g_i (the centralizer of g_i is the collection of all elements that commute with g_i), and since the centralizer is a subgroup (Judson, 185), the order of the centralizer must divide the order of the group. So the centralizer is of order 16 and hence the group. Then g_i commutes with all elements, so it is in the center, which contradicts the fact that it is in the center. hence g_i, g_j do not commute, so their commutator is a nontrivial element of the center.

We are now ready to classify groups. Since the center is of order 4, it is isomorphic to either $\mathbb{Z}_4$ or $\mathbb{Z}_2 \times \mathbb{Z}_2$. We will break these into two cases.

5.1 $Z(G) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

Since $Z(G) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, we know that each G_i contains a subgroup isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. Since all G_i are abelian, and we know that $\mathbb{Z}_8$ contains no subgroup isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$, the only possible isomorphism classes for the three G_i are $\mathbb{Z}_4 \times \mathbb{Z}_2$ or $\mathbb{Z}_2 \times \mathbb{Z}_2$. We now have four possibilities; (5.1.1) all $G_i \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, (5.1.2) two $G_i \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, (5.1.3) one $G_i \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, and (5.1.4) no $G_i \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ (so all $\cong \mathbb{Z}_4 \times \mathbb{Z}_2$). Considering the situations:

5.1.1 $G_1, G_2, G_3 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

To investigate the scenario of all $G_i \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, we need the following theorem:

Theorem 5.1.1.2 *If every nonidentity element in G has order 2, then G is abelian (Judson, 46).*

From theorem 3.4, every element is in a G_i . If all $G_i \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ then every non-identity element has order 2 (as every element is in a subgroup with nonidentity elements having order 2), and from Judson, we see that this implies that the group is abelian, so $|Z(G)| = 16 \neq 4$, so this leads to a contradiction. Hence, there are no groups with this property.

5.1.2 $G_1, G_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2, G_3 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$

We will now state conditions that we will use over and over again, so we will call them:

Conditions [1] let $g_i \in G_i, g_j \in G_j$ be elements with the property that $g_i, g_j \notin G_i \cap G_j$. Let g' be the commutator between g_j, g_i (this means $g_j g_i = g' g_i g_j$). We note that, for $g' \in Z(G)$ $(g_i g_j)^2 = g_i (g_j g_i) g_j = g_i g' g_i g_j^2 = g' g_i^2 g_j^2$.

Pick an element $r \in G_3, s \in G_1$ with conditions 1 such that $|r| = 4$ and $|s| = 2$ (we have these elements since there is a cyclic subgroup of order 4 in G_3 and a subgroup $G_1 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ that is not the center). Let z' be the commutator between s and r (we will see the reason for calling these elements r, s shortly). We note that $r^2 \in Z(G)$, as $|r^2| = 2$ and all elements of order 2 in G_3 are in the center (property of $\mathbb{Z}_4 \times \mathbb{Z}_2$). We note that $rs \in G_2$, so its order is 2 (theorem 4.4),

and thus $e = (rs)^2 = zr^2s^2 = zr^2$. Hence $z = (r^2)^{-1} = r^2$, so $sr = r^2rs = r^3s = r^{-1}s$. Thus we can form the subgroup $H = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\} \subset G$. We see that H is a subgroup as H is closed (using the commuting rule) and has inverses (elements in $\langle r \rangle$ have inverses in $\langle r \rangle$ and elements not in $\langle r \rangle$ have order 2). If we choose $z \in Z(G)$ so $z \neq e, r^2$, then $\{z, e\} = \langle z \rangle = K$ is a subgroup. We note that $H \cap K = \{e\}$, so $|HK| = |H||K|/|H \cap K| = 8 \cdot 2 / 1 = 16$, so $HK = G$, and since K is a subgroup of the center $hk = kh$ for all $k \in K, h \in H$. Hence, G is the direct internal product of $H \cong D_4$ and $K \cong \mathbb{Z}_2$, so $G \cong H \times K \cong D_4 \times \mathbb{Z}_2$. This group has representation $G = \{r^\alpha s^\beta z^\gamma : r^4 = s^2 = z^2 = e, sr = r^{-1}s, zr = rz, zs = sz\}$.

5.1.3 $G_1 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2, G_2, G_3 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$

In this case pick elements g_2, g_3 , with conditions [1], which since they are not in the center must have order 4. Let z' be the commutator between g_3 and g_2 . we know that the elements squared are in the center, so let $g_2^2 = z_2, g_3^2 = z_3$. We note that $g_2g_3 \in G_1$, so $|g_2g_3| = 2$. This means that $e = (g_2g_3)^2 = z'_2g_2^2g_3^2 = z'_2z_2z_3$. We next prove a fact about $\mathbb{Z}_2 \times \mathbb{Z}_2$:

Theorem 4.1.3.1 *let a, b, c be nonidentity elements in $\mathbb{Z}_2 \times \mathbb{Z}_2$. $abc = e$, if and only if none of the elements are equal*

Proof: ($\Leftarrow$) If $c \neq a, c \neq b$, and $c \neq e$, then the only element left is $c = ab$. Consequently, $abc = cc = c^2 = e$.

($\Rightarrow$) We will prove the contrapositive so let two of the elements be equal and we want to show $abc \neq e$. Without loss of generality assume $a = b$. then $abc = a^2c = c \neq e$, so since the contrapositive is true, the statement is true, so $abc = e$ if and only if $a \neq b \neq c$

From the above theorem, we see that $z' \neq z_2 \neq z_3$, so $z' = z_2z_3$ from the properties of $\mathbb{Z}_2 \times \mathbb{Z}_2$. Thus, we get that $g_3g_2 = (z_2z_3)g_2g_3 = z_2g_2z_3g_3 = g_2^3g_3^3 = g_2^{-1}g_3^{-1}$. Next, we note that $\langle g_2 \rangle = \{e, g_2, z_2, z_2g_2\}$ and $\langle g_3 \rangle = \{e, g_3, z_3, z_3g_3\}$, so $\langle g_2 \rangle \cap \langle g_3 \rangle = \{e\}$. Thus $|\langle g_2 \rangle \langle g_3 \rangle| = |\langle g_2 \rangle||\langle g_3 \rangle|/|\langle g_2 \rangle \cap \langle g_3 \rangle| = 4 \cdot 4 / 1 = 16$. Thus we have 16 distinct products of the form $g_2^\alpha g_3^\beta$, so group has representation $G = \{g_2^\alpha g_3^\beta : g_2^4 = g_3^4 = e, g_3g_2 = g_2^{-1}g_3^{-1}\}$ (this is called semidirect product of $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_4, G = (\mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_4$).

5.1.4 $G_1, G_2, G_3 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$

Grab $i \in G_1, j \in G_2$ with conditions [1], so since they not in the center, so $|i| = |j| = 4$. Their squares are in the center (property of $\mathbb{Z}_4 \times \mathbb{Z}_2$), so Let $i^2 = z_1, j^2 = z_2$. We note that $ij \in G_3$, so $|ij| = 4$ thus let $z_3 = (ij)^2$. This tells us that $z_3 = (ij)^2 = ijij = iz'ij^2 = z'i^2z_2 = z'z_1z_2$. by multiplying both sides by z_3z' , we get $z' = z_1z_2z_3$. Since $z' \neq e$ (see theorem 4.6) we can use the negation of theorem 4.1.3.1 (if $abc \neq e$, then at least two of the elements are equal), to state two possibilities (a) all three of z_1, z_2, z_3 are equal, or (b) two are equal.

Assume all three of z_1, z_2, z_3 are equal

In this case, we note that $z' = z_1z_2z_3 = z_1^3 = z_1$, so form the set $H = \{e, i, z_1, z_1i = i^{-1}, j, z_1j = j^{-1}, ij, z_1ij = ji = (ij)^{-1}\}$. This is a subgroup with the inverse shown (as $ij(ji) = ij^2i = iz_1i = i^2i = i^4 = e$), and the subgroup is closed with the commuting rule. Next, grab a $z \neq z_1$, and form the group $\{e, z\} = \langle z \rangle = K$. We note that $H \cap K = \{e\}$, and $|HK| = |H||K|/|H \cap K| = 8 \cdot 2 / 1 = 16$, so $HK = G$. Also, since K is a subgroup of the center, $hk = kh$ for all $h \in H, k \in K$. Then G is the direct internal product of $H \cong Q$ and $K \cong \mathbb{Z}_2$, so $G \cong H \times K \cong Q \times \mathbb{Z}_2$. This group has representation $G = \{i^\alpha j^\beta z^\gamma : i^2 = j^2, i^4 = j^4 = z^2 = e, ji = i^{-1}j, zi = iz, jz = zj\}$.

Assume two of z_1, z_2, z_3 are equal

Without loss of generality, assume $z_2 = z_3$ (we can do that, as in this case $z' = z_1 z_2 z_3 = z_1$, versus when $z_1 = z_2$, then $z' = z_1 z_2 z_3 = z_3$ or $z_1 = z_3$, in which case $z' = z_1 z_2 z_3 = z_1 z_3 z_2 = z_2$, so no matter which two of i, j , or ij have their squares equal, we always get the third as the commutator). In this case, $\langle i \rangle = \{e, i, z_1, z_1 i\}$, and $\langle j \rangle = \{e, j, z_2, z_2 j\}$. For these two subgroups $\langle i \rangle \cap \langle j \rangle = \{e\}$, and $|\langle i \rangle||\langle j \rangle| = |\langle i \rangle \langle j \rangle|/|\langle i \rangle \cap \langle j \rangle| = 4 * 4/1 = 16$, so we have sixteen product of the form $i^\alpha j^\beta$, and a commuting rule of $ji = z'ij = i^2ij = i^{-1}j$. Thus this group has a representation $G = \{i^\alpha j^\beta : i^4 = j^4 = e, ji = i^{-1}j\}$. this is called the semidirect product of $\mathbb{Z}_4$ and $\mathbb{Z}_4$, $G \cong \mathbb{Z}_4 \rtimes \mathbb{Z}_4$. We now have all groups with $Z(G) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, shown in the table below:

Name	Symbol	Representation	Center
Direct Product of D_4 and $\mathbb{Z}_2$	$D_4 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta c^\gamma : a^4 = b^2 = c^2 = e, ba = a^{-1}b, ca = ac, cb = bc\}$	$\{e, a^2, c, a^2c\}$
Semidirect product of Klein Group and $\mathbb{Z}_4$	$(\mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_4$	$\{a^\alpha b^\beta : a^4 = b^4 = e, ba = a^{-1}b^{-1}\}$	$\{e, a^2, b^2, a^2b^2\}$
Direct Product of $\mathcal{Q}$ and $\mathbb{Z}_2$	$\mathcal{Q} \times \mathbb{Z}_2$	$\{a^\alpha b^\beta c^\gamma : a^2 = b^2, a^4 = b^4 = c^2 = e, ba = a^{-1}b, ca = ac, cb = bc\}$	$\{e, a^2, c, a^2c\}$
Semidirect product of $\mathbb{Z}_4$ and $\mathbb{Z}_4$	$\mathbb{Z}_4 \rtimes \mathbb{Z}_4$	$\{a^\alpha b^\beta : a^4 = b^4 = e, ba = a^{-1}b, \}$	$\{e, a^2, b^2, a^2b^2\}$

Table 3: Non-abelian Groups of Order 16 with $Z(G) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

5.2 $Z(G) \cong \mathbb{Z}_4$

Since $Z(G) \cong \mathbb{Z}_4$, we know that each G_i contains a subgroup isomorphic to $\mathbb{Z}_4$. Since all the G_i are abelian, and we know that $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ has no subgroup isomorphic to $\mathbb{Z}_4$, so the only possible isomorphism classes are $\mathbb{Z}_8$ and $\mathbb{Z}_4 \times \mathbb{Z}_2$. We now have four possibilities, (5.2.1) all three $G_i \cong \mathbb{Z}_4 \times \mathbb{Z}_2$ (5.2.2) two $G_i \cong \mathbb{Z}_4 \times \mathbb{Z}_2$, (5.2.3) one $G_i \cong \mathbb{Z}_4 \times \mathbb{Z}_2$, and (5.2.4) no $G_i \cong \mathbb{Z}_4 \times \mathbb{Z}_2$.

5.2.1 $G_1, G_2, G_3 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$

Grab g_1, g_2 with conditions [1], and let their orders be $|g_1| = |g_2| = 2$. We note that $g_1 g_2 \in G_3$ is not in the center, so it has order 4 or 2. If we take $(g_1 g_2)^2 = z' g_1^2 g_2^2 = z'$. If $z' \in Z(G) \cong \mathbb{Z}_4$, has order 4, then $|g_i g_j| = 8$, which would imply $G_3 \cong \mathbb{Z}_8$ (not this case) so $|g_i g_j| = 4$, and $|z'| = 2$. If we grab a generator $z \in Z(G)$, (so $z^2 = z'$ and $\langle z \rangle = Z(G)$) then we see from theorem 3.4 that $G_1 = Z(G) \cup g_1 Z(G)$, so every element in G_1 looks like $z^\alpha g_1^\beta$ (where since $|g_1| = 2$ means that $\beta = 0$ (it is in $Z(G)$) or $\beta = 1$ (it is in $g_1 Z(G)$)). We next form $\langle g_2 \rangle = \{e, g_2\}$, and note $G_1 \cap \langle g_2 \rangle = \{e\}$. Thus the group $G_1 \langle g_2 \rangle$ has order $|G_1 \langle g_2 \rangle| = |G_1||\langle g_2 \rangle|/|G_1 \cap \langle g_2 \rangle| = 2 * 8/1 = 16$, so we have sixteen products of the form $z^\alpha g_1^\beta g_2^\gamma$, and a commuting rule $g_2 g_1 = z' g_1 g_2 = z^2 g_1 g_2$ so using this we can write a representation for G as $G = \{z^\alpha g_1^\beta g_2^\gamma : z^4 = g_1^2 = g_2^2 = e, g_1 z = z g_1, g_2 z = z g_2 g_2 g_1 = z^2 g_1 g_2, \}$ (this is the group of Pauli matrices).

5.2.2 $G_1, G_2, G_3 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$

As before, choose g_2, g_3 with conditions [1], and let them both be of order 2. We note that, as before $g_2g_3 \in G_1$ is not in the center, but now g_2g_3 has order 8. This means that $(g_1g_2)^2 = z'$ (from above) has order 4. We note with the following theorem that this is impossible:

Theorem 5.2.2.1 *If the commutator between any two elements is in the center, then it must be an element of order 2 (or e).*

Proof: we note that, regardless of whether the G_i 's are $\cong \mathbb{Z}_4 \times \mathbb{Z}_2$ or $\cong \mathbb{Z}_8$ when we square any element, we get an element of $Z(G)$ (as squaring an element in $\mathbb{Z}_8$ puts you in the only subgroup of order 4, which in this case is the center, and squaring an element in $\mathbb{Z}_4 \times \mathbb{Z}_2$ either gives $(0,0)$ or $(2,0)$, both of which are in all three subgroups of order 4). If we have conditions [1] with g_i, g_j , and taking $g_jg_i^2 = g_i^2g_j$. However, using our commuting rule, we get $g_jg_i^2 = z'g_i g_j g_i = z'g_i z' g_i g_j = z'^2 g_i^2 g_j$ and thus $g_i^2 g_j = z'^2 g_i^2 g_j$, and canceling the $g_i^2 g_j$ gives $e = z'^2$, so $|z'| = 2$ or $z' = e$.

We then get a contradiction if the commutator has order 4, so no group of order 16 has this property.

5.2.3 $G_1, G_2 \cong \mathbb{Z}_8, G_3 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$

Choose g_1, g_2 with conditions [1]. Since these are not in the subgroup of $G_2 \cong \mathbb{Z}_8$, of order 4 (which is $Z(G)$) they have order 8. Call $g_1^2 = z$, so $z \in Z(G)$ and $|z| = 4$. We note that we can choose a $g_2 \in G_2$ so $(|g_2| = 8)$ and $g_2^2 = z$ (you can think of g_1, g_2 as corresponding to 1 mod 8, in each group and since their intersection is $Z(G)$, squaring them will give 2 mod 8). If we call $g_3 = g_1g_2$, and let z' be the commutator of g_2 and g_1 , then $g_3 \in G_3$ (theorem 3.5) so $|g_3|$ is 4 or 2. We note that $g_3^2 = (g_1g_2)^2 = z'g_1^2g_2^2 = z'z^2$. We note that if the commutator is of order 4, then $g_3^2 = z$ or $g_3^2 = z^3$ (depending upon if $z' = z$ or $z' = z^{-1}$) either way, this would mean that $|g_3^2| = 4$ (z and z^3 have orders 4), and thus $|g_1| = 8$ (as opposed to assuming the order of the commutator is 2, I want to deduce this from the properties of the group, so if I deduce that the commutator is of order 4, I get a contradiction). This would mean $G_1 \cong \mathbb{Z}_8$, which is not the case, hence z' is of order 2, and from the properties of $Z(G) \cong \mathbb{Z}_4$, $z' = z^2$. This means that $g_3^2 = z'z^2 = z^4 = e$, so $|g_3| = 2$ and thus $\langle g_3 \rangle = \{e, g_3\}$ is a subgroup of order 2. and we note that, since $\langle g_1 \rangle = G_1$ that $G_1 \langle g_3 \rangle$ has order $|G_1 \langle g_3 \rangle| = |G_1| |\langle g_3 \rangle| / |G_1 \cap \langle g_3 \rangle| = 8 * 2 / 1 = 16$, so we have sixteen products of the form $g_1^\alpha g_2^\beta$, and since $g_3g_1 = (g_1g_2)g_1 = g_1(g_2g_1) = g_1(z'g_1g_2) = z'g_1(g_1g_2) = z'g_1g_3$, we have a commuting rule $g_3g_1 = z'g_1g_3 = z^2g_1g_3 = g_1^4g_1g_3 = g_1^5g_3$. Our group now has the representation $G = \{g_1^\alpha g_3^\beta : g_1^8 = g_3^2 = e, g_3g_1 = g_1^5g_3\}$ (this is the Isanowa or Modular Group of order 16).

5.2.4 $G_1, G_2, G_3 \cong \mathbb{Z}_8$

As before, choose g_2, g_3 with conditions [1], and with $|g_2| = 8 = |g_3|$ and picking g_2, g_3 such that $g_2^2 = z = g_3^2$. Again $g_1 = g_2g_3$ with $g_1 \in G_1$, (so $|g_1| = 8$), and $g_1^2 = (g_2g_3)^2 = z'z^2$. Note that if z' is of order 2, the $|g_1| = 2$, which contradicts the order, so $|z'| = 4$. But this contradicts theorem 4.2.1, so there are no groups with this property.

We have hence considered all cases for $Z(G) \cong \mathbb{Z}_4$, and thus all cases of $|Z(G)| = 4$. We have listed the groups of order 16 with $Z(G) \cong \mathbb{Z}_4$ in the table below.

name	symbol	representation	Center
Group of the Pauli Matrices	$SU(2)$	$\{a^\alpha b^\beta c^\gamma : a^4 = b^2 = c^2 = e, ba = ab, ca = ac, cb = a^2 bc\}$	$\{e, a, a^2, a^3\}$
Modular or Isan-owa group of order 16	M_{16}	$\{a^\alpha b^\beta : a^8 = b^2 = e, ba = a^5 b\}$	$\{e, a^2, a^4, a^6\}$

Table 4: Non-abelian groups of order 16 with centers $\mathbb{Z}_4$

6 $|Z(G)| = 2$

When the center is of order 2 (so $Z(G) \cong \mathbb{Z}_2$) then the central factor group has order $|G/Z(G)| = |G|/|Z(G)| = 16/2 = 8$, so $G/Z(G)$ is isomorphic to one of the groups of order 8, so we have 5 cases (a) $\mathbb{Z}_8$, (b) $\mathbb{Z}_4 \times \mathbb{Z}_2$, (c) $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, (d) D_4 , or (e) $\mathcal{Q}$.

(a) **assume** $G/Z(G) \cong \mathbb{Z}_8$

In this case, the central factor group is cyclic, so this implies that the group is abelian (theorem 3.1) which implies $Z(G) = G$ so $|Z(G)| = 16$, which contradicts that $|Z(G)| = 2$. Hence $G/Z(G) \not\cong \mathbb{Z}_8$.

(b) **assume** $G/Z(G) \cong \mathbb{Z}_4 \times \mathbb{Z}_2$

We note that $\mathbb{Z}_4 \times \mathbb{Z}_2$ has three subgroups, two subgroups isomorphic to $\mathbb{Z}_4$ (which has 1 subgroup $\cong \mathbb{Z}_2$) and one subgroup $\cong \mathbb{Z}_2 \times \mathbb{Z}_2$ (which has 3 subgroups $\cong \mathbb{Z}_2$). By the correspondence theorem, this means there are 3 subgroups of order 8 that contain $Z(G)$, two of which have one subgroup of order 4 that contain the center (call these G_1, G_2), and one subgroup of order 8 that contains three subgroups of order 4 with the center (call this G_3). We then prove the following theorem:

Theorem 6.1 *If $|Z(G)| = 2$ and G_i contains 1 subgroup of order 4 with the center, then G_i is either isomorphic to $\mathbb{Z}_8$ or $\mathbb{Z}_4 \times \mathbb{Z}_2$.*

Proof: We know from Theorem 3.1 that the center of G_i contains $Z(G)$. However, from the properties of the groups of order 8, we see that, for the non-abelian groups (D_4 and $\mathcal{Q}$) they have three subgroups that contain the center, so groups with this property are not abelian. Likewise, they cannot be isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$, because every subgroup of order 2 is contained in three subgroups of order 4, and not one. Furthermore, if $G_i \cong \mathbb{Z}_4 \times \mathbb{Z}_2$ then it must be in the subgroup $\cong \mathbb{Z}_2 \times \mathbb{Z}_2$, as the two subgroups $\cong \mathbb{Z}_4$ share their subgroup of order 2. The only two options are consequently $G_i \cong \mathbb{Z}_8, \mathbb{Z}_4 \times \mathbb{Z}_2$.

Therefore, since G_1, G_2 has 1 subgroup of order 4 containing the center, they are abelian. We now prove that this leads to a contradiction:

Theorem 6.2 *If G has two G_i that are abelian (could also be isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$), then the center has order at least 4.*

Proof: Let G_i, G_j be distinct and abelian. We note that $G_i G_j$ is equal to the group (as $|G_i G_j| = |G_i| |G_j| / |G_i \cap G_j| = 8 * 8 / 4 = 16$) and we note that, since they are abelian, every element in $G_i \cap G_j$ commutes with the elements in G_i and G_j . Therefore, if $g \in G_i \cap G_j$, then g commutes with every element in $G_1 G_2$ (g commutes with all $g_1 \in G_1, g_2 \in G_2$, and

an arbitrary element in G_1G_2 is g_1g_2 , thus $g(g_1g_2) = (gg_1)g_2 = g_1gg_2 = (g_1g_2)g$. Since it commutes with every element in the group, it is in the center. This means that every element in $G_i \cap G_j$ is in the center, and since there are four elements in $G_i \cap G_j$, there must be four elements in the center.

Since we have two abelian groups (G_1, G_2) , the center must be of at least order 4, but this would contradict that the center is of order 2. Hence $G/Z(G) \not\cong \mathbb{Z}_4 \times \mathbb{Z}_2$.

(c) **assume** $G/Z(G) \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

To begin, we note that since every element in the factor group has order 2, $Z(G) = (gZ(G))^2 = g^2Z(G)$, so every element squared is in the center. We note that since the group is not abelian, there exists 1 element g that does not have $g^2 = e$ (theorem 4.1.1.1). Since $g^2 \in Z(G)$, however, we note that $g^2 = z$. We note that $\langle g \rangle = \{e, g, g^2 = z, g^3 = zg\}$ is a subgroup containing the center, so by the correspondence theorem there is a factor group of order 2 in the center (in this case, $\langle gZ(G) \rangle = \{Z(G), gZ(G)\}$). We note that this group is contained in the three subgroups of order 4 (each of the the subgroup of order 2 is contained in 3 subgroups in $G/Z(G) \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$). If we call these G_1, G_2, G_3 , then we note at least two of these are nonabelian (if two are abelian it implies a center of order 4, see Theorem 5.2). Without loss of generality assume G_2, G_3 are nonabelian. Then we note that if we let $g_2 \in G_2, g_3 \in G_3$ which are not in their intersection (which in this case is $\langle g \rangle$). then we note that $g_2g_3 \in G_1$ and is not in $\langle g \rangle$. then we note that, since $g \notin Z(G_2), Z(G_3)$ (which are of order 2 so equal the center) when we commute we pick up a z , $g_2g = zgg_2$ (same for g_3). If we then take $(g_2g_3)g$, we see that $(g_2g_3)g = g_2(g_3g) = g_2(zgg_3) = z(g_2g)g_3 = z(zgg_2)g_3 = z^2g(g_2g_3) = g(g_2g_3)$, so g commutes with $g_2g_3 \in G_1$, and we note that g commutes with $\langle g \rangle \cup (g_2g_3)\langle g \rangle$, which are eight elements G_1 hence it equals G_1 , hence $g_1 \in Z(G_1)$, and now the center of G_1 has at least three elements (g, e, z) it has to be abelian. This center has no elements of order 8, as these do not have $g^2 = z$, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ has no elements of order 2, so $G_1 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$.

We next let $g_1 \in G_1, g_1 \notin \langle g \rangle$. If $h_2 \in G_2$ commutes with g_1 , then $g_1^\alpha h_2^\beta g^\gamma$ is an abelian group with eight elements, as everything commutes with everything else, and has eight elements by two options for each greek letter (if either an element of order 8, you can use that g_1^2 and/or h_2^2 is equal to z) then $G_i = \{g_1^\alpha h_2^\beta z^\gamma\}$ is an abelian group, we have two abelian groups of order 8, which implies that $|Z(G)| \geq 4$ which contradicts that $|Z(G)| = 2$. Thus h_2 cannot commute with any elements in G_1 that are not in $\langle g \rangle$. Thus g_2 does not commute with g_1 , but we note that g_1g is not in $\langle g \rangle$ (as then $g_1 = g^{-1}g^n = g^{n-1}$ is in $\langle g \rangle$). But $gg_1g_2 = g(zg_2g_1) = z(gg_2)g_1 = z^2g_2gg_1 = g_2(gg_1)$ has g_2 commuting with gg_1 , which would contradict the fact that it does not commute, which if it did commute we could construct a second abelian group and thus $|Z(G)| \geq 4$, which contradicts that $|Z(G)| = 2$.

(d) **assume** $G/Z(G) \cong \mathcal{Q}$

We note from the properties of $\mathcal{Q}$ that the factor group has three subgroups of order 4, each with 1 subgroup of order 2. By the Correspondence Theorem, this means there are 3 subgroups of order 8 in G , each with 1 subgroup of order 4 (call these G_i). By Theorem 6.1, all three of these are abelian, and since there are 2 abelian G_i , the center must have order 4, contradicting that the center has order 2. Hence no groups of order 16 have this property.

(e) **assume** $G/Z(G) \cong D_4$

In this case we note that the factor group has 1 subgroup of order 4 with 1 subgroup of order 2,

and 2 subgroups of order 4 with three subgroups of order 2. By the Correspondence Theorem, we have 1 subgroup of order 8 with 1 subgroup of order 4 containing the center (call this G_1 and from Theorem 5.1, $G_1 \cong \mathbb{Z}_8$ or $\mathbb{Z}_4 \times \mathbb{Z}_2$), and two subgroups of order 8 with three subgroups containing the center (denote these G_2, G_3). We note that both G_2, G_3 are nonabelian, as if they were abelian then by theorem 5.2, $Z(G)$ would have order 4. Before we determine the nature of these subgroups we need to determine how elements commute. This means finding out the commutator subgroup:

Definition the commutator subgroup, G' , is defined as $G' = \langle ghg^{-1}h^{-1} : g, h \in G \rangle$ (Judson, 202).

Note that $Z(G) \subset Z(G_2)$, and in these nonabelian groups of order 8, the center is of order 2, hence $Z(G) = Z(G_2)$. Also, based on the properties of these groups, the center is contained in the commutator (for $G_2 \cong D_4$, then $z = r^2$ is the nontrivial element of the center, and $z = r^2 = srs^{-1}r^{-1}$, so z an element of G' . for $G_2 \cong Q$, then $z = -1 = i^2$ is a nontrivial element of the center, and $z = -1 = i^2 = iji^{-1}j^{-1}$ is in G'). Let $Z(G), g'Z(G)$ be the center of $G/Z(G)$ (so g' is equivalent to r^2 in D_4). If we pick $g_1Z(G), g_2Z(G)$ in the factor group so they do not commute, then from the properties of D_4 , they pick up an element of the center when they commute $g_1g_2Z(G) = (g_1Z(G))(g_2Z(G)) = (g'Z(G))(g_2Z(G))(g_1Z(G)) = g'g_1g_2Z(G)$, so $g_1g_2 = g'z_0g_2g_1$, for some z_0 in the center, hence g', zg' are in the commutator subgroup (as regardless of z_0 the fact that the commutator is a subgroup means we can multiply by its inverse in the center (which is in the commutator) and get g' , and from this zg'). Note that these four elements form a subgroup, and we have no other generators for G' (assume $gh = chg$ with c in the commutator. Then $gZ(G)hZ(G) = ghZ(G) = chgZ(G) = cZ(G)hZ(G)gZ(G)$ and in the factor group, we either pick up a $g'Z(G) = \{g', zg'\}$ or $Z(G) = \{e, z\}$, so commuting picks up one of these four elements, which are the four elements in the center). This we get no more generators, thus $G' = \{e, z, g', zg'\}$ is the commutator. We note that G' corresponds to the center of the factor group, and the center is contained in the four subgroups of order 4, hence by the correspondence theorem this subgroup is in all three subgroups of order 8 containing the center ($G' \subset G_1, G_2, G_3$), and note that $|G_i \cap G_j| = 4$, so the intersection of two G_i is G' .

We now need to determine the isomorphism classes of G_1, G_2, G_3 . We will determine the structure of G_1 (the abelian one).

Theorem 6.3 if $G/Z(G) \cong D_4$ then one of the G_i is cyclic

Proof: We already know that $G_1 \cong \mathbb{Z}_8$ or $\mathbb{Z}_4 \times \mathbb{Z}_2$, so we will do a proof by contradiction. Assume $G_1 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$. Note from theorem 6.1 that this means that the one subgroup of order 4 containing the center is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. We know G_2, G_3 are nonabelian, so they are isomorphic to Q or D_4 . However, neither are isomorphic Q , as the only subgroups of order 4 in Q are cyclic, so this would mean that $G' = G_1 \cap G_i \cong \mathbb{Z}_4$ is cyclic, which contradicts that it is $\mathbb{Z}_2 \times \mathbb{Z}_2$. This leaves both $G_2, G_3 \cong D_4$. since there are five elements of order 2 in D_4 , we can pick g_2 such that $|g_2| = 2$. We note that, for $g_1 \in G_1$, (and commutator g' between g_2 and g_1). Then $g_1 = g_2^2g_1 = g_2g'g_1g_2 = zg'g_2g_1g_2 = zg'g'g_1g_2^2 = zg'^2g_1$ (we note that $g', g_2 \in G_2$ commute with an element of center from the properties of groups of order 8). Canceling out g_1 gives $e = g'^2z$ and thus $g'^2 = z^{-1} = z$ (as inverse of element of order 2 is itself) thus $|g'| = 4$. But this contradicts that the group of order 4 containing the center is $\cong \mathbb{Z}_4 \times \mathbb{Z}_2$, hence this is a contradiction, so there is no way for $G_1 \cong \mathbb{Z}_4 \times \mathbb{Z}_2$.

Now that we know $G_1 \cong \mathbb{Z}_8$, we have three scenarios, (5.0.1) $G_2, G_3 \cong Q$, (5.0.2) $G_2 \cong Q, G_3 \cong D_4$, or (5.0.3) $G_2, G_3 \cong D_4$.

5.0.1 $G_2, G_3 \cong \mathcal{Q}$

Grab g_1, g_2 with conditions [1]. Note that, since $G_1 \cong \mathbb{Z}_8$ shares a subgroup of order 4 with $G_2 \cong \mathcal{Q}$, we have $(g_1^4 = g'^2 = g_2^2 = z)$. Also, $g_1 g_2 \in G_3$, so $z = (g_1 g_2)^2 = g_1 g_2 g_1 g_2 = g_1 g' g_1 g_2 g_2 = g' g_1^2 z$, so $e = g' g_1^2$, so $g' = (g_1^2)^{-1} = g_1^{-2} = g_1^6$. If we take $\langle g_1 \rangle = G_1$, and $\langle g_2 \rangle = \{e, g_2, z, z g_2\}$ (so $G_1 \cap \langle g_2 \rangle = \{e, z\}$), and form $G_1 \langle g_2 \rangle$, then this has order $|G_1 \langle g_2 \rangle| = |G_2| |\langle g_2 \rangle| / |G_1 \cap \langle g_2 \rangle| = 8 * 4 / 2 = 16$, so there are 16 elements of the form $g_1^\alpha g_2^\beta$ and commuting rule $g_2 g_1 = g' g_1 g_2 = g_1^6 g_1 g_2 = g_1^7 g_2 = g_1^{-1} g_2$. So our group has the representation $G = \{g_1^\alpha g_2^\beta : g_1^4 = g_2^2, g_1^8 = g_2^4 = e, g_2 g_1 = g_1^{-1} g_2\}$, This is the generalized quaternions, (also dicyclic group of degree 4, Dic_4).

5.0.2 $G_2 \cong D_4, G_3 \cong \mathcal{Q}$

Grab g_1, g_3 with conditions [1], and note that $g_1 g_3 \in G_2 \cong D_4$ is not in the cyclic subgroup of order 4, so has order 2, $(g_1 g_3)^2 = e$. This means that $e = (g_1 g_3)^2 = g_1 g_3 g_1 g_3 = g_1 g' g_1 g_3^2 = g_1^2 g' (g_1^4) = g' g_1^6$, so $g' = (g_1^6)^{-1} = g_1^{-6} = g_1^2$. If we let $g_2 = g_1 g_3$, then to determine commutator between g_2 and g_1 , note $g_2 g_1 = g_1 g_3 g_1 = g_1 g' g_1 g_3 = g_1^2 g_1 (g_1 g_3) = g_1^3 g_2$, and we note that since $|g_2| = 2$, $\langle g_2 \rangle = \{e, g_2\}$, and $\langle g_1 \rangle = G_1$, and $|G_1 \langle g_2 \rangle| = |G_1| |\langle g_2 \rangle| / |G_1 \cap \langle g_2 \rangle| = 8 * 2 / 1 = 16$, so sixteen elements of the form $g_1^\alpha g_2^\beta$, so using the orders and commuting rules we get a group representation of $G = \{g_1^\alpha g_2^\beta : g_1^8 = g_2^2 = e, g_2 g_1 = g_1^3 g_2\}$. This is the semidihedral group of degree 2, SD_2 .

5.0.3 $G_2, G_3 \cong D_4$

Grab $g_1 \in G_1, g_2 \in G_2$ with conditions [2]. Let g' be the commutator between them $g_2 g_1 = g' g_1 g_2$, and $|g_2| = 2$. Since $g_1 g_2 \in G_3 \cong D_4$, and not in cyclic group of order 4, $|g_1 g_2| = 2$. We note that $e = (g_1 g_2)^2 = g_1 g_2 g_1 g_2 = g_1 g' g_1 g_2^2 = g' g_1^2$, so $g' = (g_1^2)^{-1} = g_1^{-2} = g_1^6$. Next, we note that $\langle g_1 \rangle = G_1$, and $\langle g_2 \rangle = \{e, g_2\}$, so $|G_1 \langle g_2 \rangle| = |G_1| |\langle g_2 \rangle| / |G_1 \cap \langle g_2 \rangle| = 8 * 2 / 1 = 16$, so sixteen elements of the form $g_1^\alpha g_2^\beta$ with commuting rule $g_2 g_1 = g' g_1 g_2 = g_1^6 g_1 g_2 = g_1^7 g_2 = g_1^{-1} g_2$. This gives a representation of $G = \{g_1^\alpha g_2^\beta : g_1^8 = g_2^2 = e, g_2 g_1 = g_1^{-1} g_2\}$ (this is the dihedral group of degree 8, D_8).

Thus we have determined the 14 groups of order 16, which are all collected in a table in the appendix

name	symbol	representation	Center
Dicyclic Group of Degree 4	Dic_4	$\{a^\alpha b^\beta : a^4 = b^2 a^8 = b^4 = e, ba = a^{-1}b\}$	$\{e, a^4\}$
Semidihedral group of degree 2	SD_2	$\{a^\alpha b^\beta : a^8 = b^2 = e, ba = a^3b\}$	$\{e, a^4\}$
Dihedral group of degree 8	D_8	$\{a^\alpha b^\beta : a^8 = b^2 = e, ba = a^{-1}b\}$	$\{e, a^4\}$

Table 5: Groups of order 16 with $Z(G) \cong \mathbb{Z}_2$

Thus we have now determined all 14 groups of order 16, which are collected in the appendix.

7 Resources

References

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Appendix 1: The representations of the groups of order 16

name	symbol	representation	Center
Integers mod 16	$\mathbb{Z}_{16}$	$\{a^\alpha : a^{16} = e\}$	Abelian
Direct Product of $\mathbb{Z}_8$ and $\mathbb{Z}_2$	$\mathbb{Z}_8 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta : a^8 = b^2 = e, ba = ab\}$	Abelian
Direct Product of $\mathbb{Z}_4$ and $\mathbb{Z}_4$	$\mathbb{Z}_4 \times \mathbb{Z}_4$	$\{a^\alpha b^\beta : a^4 = b^4 = e, ba = ab\}$	Abelian
Direct Product of $\mathbb{Z}_4$, $\mathbb{Z}_2$ and $\mathbb{Z}_2$	$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta c^\gamma : a^4 = b^2 = c^2 = e, ba = ab, ca = ac, cb = bc\}$	Abelian
Direct product of $\mathbb{Z}_2$, $\mathbb{Z}_2$, $\mathbb{Z}_2$ and $\mathbb{Z}_2$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta c^\gamma d^\delta : a^2 = b^2 = c^2 = d^2 = e, ba = ab, ca = ac, da = ad, cb = bc, db = bd, dc = cd\}$	Abelian
Direct Product of D_4 and $\mathbb{Z}_2$	$D_4 \times \mathbb{Z}_2$	$\{a^\alpha b^\beta c^\gamma : a^4 = b^2 = c^2 = e, ba = a^{-1}b, ca = ac, cb = bc\}$	$\{e, a^2, c, a^2c\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$
Semidirect product of Klein Group and $\mathbb{Z}_4$	$(\mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_4$	$\{a^\alpha b^\beta : a^4 = b^4 = e, ba = a^{-1}b^{-1}\}$	$\{e, a^2, b^2, a^2b^2\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$
Direct Product of $\mathcal{Q}$ and $\mathbb{Z}_2$	$\mathcal{Q} \times \mathbb{Z}_2$	$\{a^\alpha b^\beta c^\gamma : a^2 = b^2, a^4 = b^4 = c^2 = e, ba = a^{-1}b, ca = ac, cb = bc\}$	$\{e, a^2, c, a^2c\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$
Semidirect product of $\mathbb{Z}_4$ and $\mathbb{Z}_4$	$\mathbb{Z}_4 \rtimes \mathbb{Z}_4$	$\{a^\alpha b^\beta : a^4 = b^4 = e, ba = a^{-1}b\}$	$\{e, a^2, b^2, a^2b^2\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$
Group of the Pauli Matrices	$SU(2)$	$\{a^\alpha b^\beta c^\gamma : a^4 = b^2 = c^2 = e, ba = ab, ca = ac, cb = a^2bc\}$	$\{e, a, a^2, a^3\} \cong \mathbb{Z}_4$
Modular or Isanowa group of order 16	M_{16}	$\{a^\alpha b^\beta : a^8 = b^2 = e, ba = a^5b\}$	$\{e, a^2, a^4, a^6\} \cong \mathbb{Z}_4$
Dicyclic Group of Degree 4	Dic_4	$\{a^\alpha b^\beta : a^4 = b^2a^8 = b^4 = e, ba = a^{-1}b\}$	$\{e, a^4\} \cong \mathbb{Z}_2$
Semidihedral group of degree 2	SD_2	$\{a^\alpha b^\beta : a^8 = b^2 = e, ba = a^3b\}$	$\{e, a^4\} \cong \mathbb{Z}_2$
Dihedral group of degree 8	D_8	$\{a^\alpha b^\beta : a^8 = b^2 = e, ba = a^{-1}b\}$	$\{e, a^4\} \cong \mathbb{Z}_2$